# Germans Savcisens, PhD

*Postdoctoral Research Associate, Northeastern University*

*Boston, MA, USA* · *Portfolio* · *LinkedIn* · *GitHub* · *Google Scholar* · *ORCID* · *Semantic Scholar* · *Scopus* · *Web of Science*

## PROFILE

Postdoctoral researcher (PhD, Applied Mathematics and Computer Science) specializing in statistical and machine-learning modeling of registry-scale health & behavior data, and in the robustness, and stability assessment of AI systems. Investigated transformer-based systems for nation-scale health/socioeconomic trajectory prediction (life2vec, trained on national registry and electronic health records), veracity probing of LLMs and multi-agent LLM simulation (CoevolveSim), with end-to-end Python experience from data extraction and cleaning to modeling and interpretability analysis. Works across multidisciplinary teams and mentors graduate researchers in interpretable ML and AI auditing.

**Areas of interest:** population-scale health & behavior modeling, explainable AI & AI auditing, bias & robustness assessment, transformer architectures, agent-based simulation, evaluation of predictive models.

## EDUCATION

**Technical University of Denmark** *2020–2024*
*PhD, Applied Mathematics and Computer Science* *Copenhagen, Denmark*

- Dissertation: *Life Trajectories as Symbolic Language: Exploring Human Behaviour with Language Models*
- Committee: Søren Hauberg (chair), Mads Nielsen, Arkadiusz Stopczynski

**Technical University of Denmark** *2018–2020*
*MSc, Human-Centered Artificial Intelligence* *Copenhagen, Denmark*

- Thesis (Collaboration with Novo Nordisk): *Analyzing Health Data Records Using Neural Networks*
- Thesis Advisors: Sune Lehmann (DTU) and Adam Lenart (Novo Nordisk)

**Aalto University** *2021*
*Visiting Student, Digital Ethics & Statistical Natural Language Processing* *Helsinki, Finland*

**Aalborg University** *2013–2016*
*BSc, Human-Computer Interaction (Medialogy)* *Copenhagen, Denmark*

## RESEARCH EXPERIENCE

**Northeastern University** *May 2024–present*
***Postdoctoral Research Associate*** | *Mentor: Tina Eliassi-Rad* *Boston, USA*

- Developed LLM probing frameworks for knowledge representation and belief stability [sAwMIL].
- Designed a multi-agent framework (using R and Python) to study opinion dynamics in networks of LLM agents [CoevolveSim].
- Supervised graduate researchers in interpretable machine learning and LLM auditing; coordinated research on multi-agent interactions.
- Presented projects at the NeurIPS Mechanistic Interpretability Workshop, IC2S2, and NEMI Workshop.

**Technical University of Denmark & University of Copenhagen** *Sep 2020–Apr 2024*
***PhD Student & Research Assistant*** | *Advisors: Sune Lehmann, Lars Kai Hansen* *Copenhagen, Denmark*

- Led modeling of nation-scale socioeconomic and health trajectories using transformer-based models [life2vec].
- Developed and evaluated life-outcome prediction models, with emphasis on interpretability.
- Taught and developed course materials for a graduate-level course in Social Network Science.
- Invited talks at MIT, University of Oxford, NordicAI and Complexity Science Hub.

**Statistics Denmark, Data Science Lab** *Sep 2020–Apr 2024*
***Guest Researcher*** *Copenhagen, Denmark*

- Collaborated with economists and social scientists to conduct quantitative and qualitative analyses of demographic and socioeconomic patterns in observational registry data.
- Led modeling of large-scale longitudinal datasets.

**Network Science Institute, Northeastern University** *Sep 2022–Feb 2023*
***Visiting PhD Student*** *Boston, USA*

- Analyzed ethical and societal risks of deep learning applied to socioeconomic data.
- Investigated methods for enhancing transparency and fairness in algorithmic systems.

**Novo Nordisk, R&D (Real World Evidence)** *Sep 2019–Apr 2020*
***Researcher (Student Assistant)*** *Copenhagen, Denmark*

- Conducted research on predictive models for medical adherence using large-scale Electronic Health Records (EHRs), addressing missing data and selection bias in the observational cohort.
- Developed hierarchical attention-based neural models for health outcome prediction.

- Led preprocessing and integration of behavioral and clinical data sources.

## PUBLICATIONS

### PEER REVIEWED & PUBLISHED

**G. Savcisens** & T. Eliassi-Rad (2026). The Trilemma of Truth in Large Language Models. ***To appear*** *in Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. [arXiv] [Code] [NeurIPS Poster]

S. Dies, C. Maynard, **G. Savcisens** & T. Eliassi-Rad (2026). Epistemic Familiarity is Associated With Belief Stability in Large Language Models. ***To appear*** *in Findings of the 2026 Conference on Empirical Methods in Natural Language Processing*. [arXiv]

Z. Shafi, **G. Savcisens** & T. Eliassi-Rad (2025). REGE: A Method for Incorporating Uncertainty in Graph Embeddings. *SIAM International Conference on Data Mining*. [DOI]

**G. Savcisens** (2025). Large language models act as if they are part of a group. *Nature Computational Science*. [DOI]

E. F. Fernández & **G. Savcisens** (2024). Quantifying Western Discourses about Technology: An Analysis of Press Coverage using a Dataset of Multilingual Newspapers. *Digital Humanities in the Nordic and Baltic Countries*. [DOI]

E. Denove, E. Michelet, **G. Savcisens** & E. F. Fernández (2024). An Industrial West? A Mixed-Methods Analysis of Newspapers Discourses about Technology over One Hundred and Ten Years (1830-1940). *Journal of Data Mining & Digital Humanities*. [DOI]

**G. Savcisens**, T. Eliassi-Rad, L. K. Hansen, L. H. Mortensen, L. Lilleholt, A. Rogers, I. Zettler & S. Lehmann (2024). Using Sequences of Life-events to Predict Human Lives. *Nature Computational Science*. [DOI] [Official Page]

E. F. Fernández & **G. Savcisens** (2023). A Sustainable West? Analyzing Clusters of Public Opinion in Sustainability Western Discourses in a Collection of Multilingual Newspapers (1999-2018). *Digital Humanities in the Nordic and Baltic Countries*. [DOI]

### PREPRINTS

M. Laber, Z. Shafi, **G. Savcisens**, B. Klein, M. Chinazzi, S. V. Scarpino, A.-L. Barabási & T. Eliassi-Rad (2026). Scaling of Capability and Efficiency at Inference Time in Large Reasoning Models. *arXiv:2609.27166*. [arXiv]

E. Sivak, …, **G. Savcisens**, et al. (112 authors) (2026). Births are difficult to predict even with rich survey and full-population register data. *arXiv:2609.01194*. [arXiv]

**G. Savcisens**, S. Dies, C. Maynard & T. Eliassi-Rad (2026). Belief Coevolution in a Social Network of Generalist and Specialist Large Language Models. *arXiv:2607.27512*. [arXiv] [Code] [NetSci Poster]

### BLOG POSTS

**G. Savcisens** (2025). Is Life2vec a Mess? What's Wrong with AI Death Calculators? *savcisens.com*. [Source]

### THESES

**G. Savcisens** (2023). Life Trajectories as Symbolic Language: Exploring Human Behaviour with Language Models. *PhD Thesis (Technical University of Denmark, Cognitive Systems)*. [DTU]

**G. Savcisens** (2020). Analyzing Health Data Records Using Neural Networks: Predicting Diabetes-Related Clinical Outcomes. *Master Thesis (Technical University of Denmark)*. [DTU]

## TEACHING & MENTORING

**Northeastern University** *2024–2026*
***Guest Lecturer*** | *Foundations of Artificial Intelligence (2026); Machine Learning with Graphs (2024–2025)*

**IT University of Copenhagen** *2022–2023*
***Principal Lecturer & Examiner*** | *Algorithmic Fairness, Accountability and Ethics (2023)* ***Assistant Lecturer*** | *Algorithmic Fairness, Accountability and Ethics (2022)*

**Technical University of Denmark** *2019–2021*
***Teaching Assistant*** | *Advanced Machine Learning (2021); Social Graphs and Interactions (2019–2020); Social Data Analysis and Visualization (2019–2020)*

***Mentoring & Supervision***

Co-supervised 5 student projects (4 MSc theses, 1 term project), Technical University of Denmark (2021–2024).

## SELECTED TALKS & PRESENTATIONS

**The Trilemma of Truth in Large Language Models** — *Oct 2026*
*Conference on Empirical Methods in Natural Language Processing (Poster)* — *Budapest, Hungary*

**Transformers in Social Science: From Human Lives to Machine Minds** — *Oct 2026*
*Human Nature Lab, Yale University (Invited Speaker)* — *New Haven, USA*

**Interpretability or Guesswork: What Happens When We Open Up Large Language Models** — *Sep 2026*
*International Conference on Social Computing, University of Oxford (Keynote Speaker)* — *Oxford, UK*

**Transformer Architectures for Computational Social Science [Workshop Page]** — *Sep 2026*
*International Conference on Social Computing, University of Oxford (Workshop Lead)* — *Oxford, UK*

**Belief Coevolution in a Social Network of Generalist and Specialist Large Language Models** — *Aug 2026*
*New England Mechanistic Interpretability Workshop, Boston University (Poster)* — *Boston, USA*

**Belief Coevolution in a Social Network of LLMs** — *Jun 2026*
*International Conference on Network Science, Northeastern University (Poster & Lightning Talk)* — *Boston, USA*

**Formal Epistemology of Large Language Models** — *Apr 2026*
*Social Complexity Lab, University of Copenhagen (Invited Speaker)* — *Copenhagen, Denmark*

**From Human Lives to Machine Minds [Slides]** — *Apr 2026*
*Rising Stars in Computational & Data Science, Santa Fe Institute (Talk)* — *Santa Fe, USA*

**The Trilemma of Truth in LLMs** — *Dec 2025*
*NeurIPS Mechanistic Interpretability Workshop (Poster)* — *San Diego, USA*

**Life2vec: Foundation Models for Registry Data [Slides]** — *Nov 2025*
*Networks for Healthy Aging Workshop, Complexity Science Hub (Invited Speaker)* — *Vienna, Austria*

**From Life-Courses to Representations: Foundation Models for Registry Data** — *Sep 2025*
*SWECOV workshop, Lund University (Keynote Speaker)* — *Stockholm, Sweden*

**The Trilemma of Truth in Large Language Models** — *Aug 2025*
*New England Mechanistic Interpretability Workshop, Northeastern University (Poster)* — *Boston, USA*

**Improving Probes that Track Veracity in LLMs** — *Jul 2025*
*Conference on Computational Social Science (IC2S2), Linköping University (Poster)* — *Norrköping, Sweden*

**Workshop on the '(Un)predictability in the Social Sciences'** — *Feb 2025*
*University of Helsinki (Invited Speaker)* — *Helsinki, Finland*

**Seminar Talk on LLMs for Health and Labor data** — *Jan 2025*
*Max Planck Institute (Invited Speaker)* — *Rostock, Germany*

**Life2vec: Life Trajectories in High-dimensional Spaces [Recording]** — *Nov 2024*
*Computational Social Science Lab at the Chinese University of Hong Kong (Invited Speaker)* — *Hong Kong, PRC (Remote)*

**Life2vec: Life Trajectories in High-dimensional Spaces** — *Apr 2024*
*GenLife (Invited Speaker)* — *Copenhagen, Denmark*

**Seminar Talk** — *Jan 2024*
*Centre for Demographic Science (University of Oxford) (Invited Speaker)* — *Oxford, UK (Remote)*

**Using Life-sequences to Predict Human Lives** — *Sep 2023*
*ODISSEI Lecture (Lecture)* — *Netherlands (Remote)*

**Danish Registry Data and Life Trajectories** — *Oct 2022*
*Human Dynamics Lab (MIT) (Invited Speaker)* — *Boston, US*

## PROFESSIONAL SERVICE

**Program Committee:** International Symposium on Complex Systems 2027, AAAI 2027 (Senior Program Committee, Special Track on AI Alignment).

**Session Chair:** Conference on Empirical Methods in Natural Language Processing (EMNLP) 2026, International Conference on Network Science (NetSci) 2026 (ML & AI track).

**Journals (reviewer):** Artificial Intelligence Review, EPJ Data Science, Humanities and Social Sciences Communications, Nature Computational Science, npj Complexity, Science Advances, Scientific Reports.

**Conferences & Workshops (reviewer):** Workshop on NLP and Computational Social Science 2026, NeurIPS AXIOM Workshop 2026, NeurIPS Mechanistic Interpretability Workshop 2025, Conference on Complex Networks and Their Applications 2025, IC2S2 2025, Digital Humanities in the Nordic and Baltic Countries 2025, EMNLP 2024.

**Ethics Reviewer:** NeurIPS 2023–2026.

## AWARDS & GRANTS

**Rising Stars in Computational and Data Sciences 2026** — *2026*
*Santa Fe Institute* — *USA*

**Travel Grants** — *2022*
*Otto Mønsted Fond, STIBO Foundation, William Demant Fonden* — *Denmark*